

Iteration 1

[0]	[1]	[2]	[3]	[4]	[5]
30	20	60	50	10	40

Pos pos+1

30 > 20 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

Pos pos+1

30 > 60 ? no

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

Pos pos+1

60 > 50 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	60	10	40

Pos pos+1

60 > 10 ? Yes

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	60	40

Pos pos+1

60 > 40 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Iteration/cycle/pass : 2

[0] > [5]

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Pos pos+1

20 > 30 ? no

[1] > [5]

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Pos pos+1

30 > 50 ? no

[2] > [5]

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Pos pos+1

50 > 10 ? Yes
swap

[3] > [5]

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	50	40	60

Pos pos+1

50 > 40 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	40	50	60

Iteration 3

[0] > [1]

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	40	50	60

20 > 30 ? no

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	40	50	60

Pos pos+1

30 > 10 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	10	30	40	50	60

Pos pos+1

30 > 40 ? no

[0]	[1]	[2]	[3]	[4]	[5]
20	10	30	40	50	60

Iterations 4

[0]	[1]	[2]	[3]	[4]	[5]
20	10	30	40	50	60

Pos pos+1

20 > 10 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

Pos pos+1

20 > 30 ? no

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

Iteration no : 5

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

Pos pos+1

Arr[pos] > arr[pos+1]
no

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

No. of Iterations : n-1

$$6-1 = 5$$

No of comparisons : (n-1) + (n-2) + (n-3).... 1

$$5 + 4 + 3 + 2 + 1 = 15$$

1. Worst and average case time complexity = $O(n^2)$

2. Best case time complexity : $O(n)$

Total Space comp = input space : $O(n)$ + aux space : $O(1)$

$O(n+1)$:

$O(n)$

Space complexity : $O(1)$:Aux space